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A comprehensive qualitative analysis of patient dialogue summarization using large language models applied to noisy, informal, non-English real-world data

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This study evaluates the ability of Large Language Models (LLMs) to summarize real-world dialogues between patients and the healthcare team of an e-health company that provides digital healthcare services, primarily communicating via WhatsApp. The team needs quick access to patient information to deliver accurate and personalized responses. Summarizing past messages is the approach examined here, aiming for concise, non-redundant, and truthful summaries that capture the main dialogue characteristics despite facing (real-world) noisy and informal content in an under-represented language - Portuguese. To do so, we collected an anonymized Portuguese dataset of WhatsApp messages exchanged between patients and the healthcare team. Dialogue quality was assessed for size, readability, and correctness before generating summaries with LLaMA3 and Qwen2 using specific prompts. Volunteers evaluated these summaries on coverage, relevance, redundancy, and veracity using a 5-point Likert scale. Our qualitative and quantitative experimental results indicate that LLMs can produce effective summaries of dialogues between patients and healthcare teams, even when faced with low-quality data in an underrepresented language. This is a surprising result due to the challenging scenario. Among the tested LLMs, LLaMA3 demonstrated a slight edge over QWen2 in coverage and veracity among the evaluated methods. Our results demonstrate a potential to build real-world practical services to assist healthcare professionals in responding to patient messages with agility, clarity, and cohesion, enhancing both communication efficiency and patient satisfaction. Ultimately, the advocated approach could significantly improve the landscape of online healthcare communication, particularly in resource-constrained settings like Brazil, where access to primary care is limited.

The increasing use of electronic health records as well as of (instant) messaging platforms such as WhatsApp and Messenger has expanded the use of e-health portals to exchange messages between patients and healthcare teams¹. The management of such an increasing volume of messages is a challenge. Indeed, primary care physicians are estimated to spend around 1.5 hours daily analyzing approximately 150 patients' messages¹. To respond effectively, health professionals must understand the message's context by reviewing previous ones from the same patient, as well as any other relevant patient information available to them.

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In Brazil, where 34% of the population lacks access to primary care, it is estimated that achieving 100% coverage would require 236,900 healthcare professionals at a cost of BRL 22.9 billion/year². An approach to scaling such services would be through a Digital Primary Care model, such as that offered by Ana Health (<https://www.anahealth.com.br/>), an e-health company that provides digital healthcare services, primarily communicating via WhatsApp, in order to scale patients. Ana Health's healthcare professionals primarily interact with the population through a specific messaging system – WhatsApp. In this interaction model, the challenge of processing and comprehending a large volume of messages remains, as it is essential to respond to each patient individually.

In more detail, at Ana Health, when a healthcare team member receives a new message from a patient, to adequately respond, the team member shall comprehend the full context, review previous messages exchanged with the patient, and understand the current message. Only after that can the healthcare team member craft a response aimed at maintaining patient engagement. For the patient to receive the response efficiently and effectively, it is essential to quickly gather contextual information in order to provide accurate, up-to-date, and personalized responses.

An approach to assist the healthcare team in understanding the context of patient messages, which we advocate and analyze in this article, is to apply summarization techniques to prior messages exchanged between the patient and the healthcare team. Such a summarization, if successful, can also support other services the healthcare team offers to the patient, such as psychotherapy, chronic disease monitoring, and drug prescription.

Previous work has developed and evaluated summarization techniques^{3–5}, including the summarization of clinical data⁴, exploiting techniques ranging from rule-based approaches to the use of Large Language Models (LLMs)⁶. However, these techniques are rarely evaluated in real environments using real data⁷. Our work tries to fulfill this literature gap. That is, our objective is to evaluate the ability of LLMs to generate useful summaries for assisting a healthcare team in understanding a patient context using real and anonymized health data and to conduct a thorough evaluation of these summaries with the support of specialists, particularly volunteers from a healthcare team.

Recent advances in LLMs based on transformers have helped in several tasks related to natural language processing, including summarization⁶. In⁸, the authors state that LLMs can assist in several healthcare tasks, such as pre-consultation, diagnosis, and management, with proper development and supervision. Despite that, some concerns and challenges still remain in the use of LLMs in healthcare, including^{8–10}: (1) data privacy – ensuring that real patient data will not be leaked and used by the owners of the LLMs for other purposes; (2) credibility and correctness of information – LLMs can perpetuate incorrect information or produce “hallucination”; and (3) biased data – LLMs are commonly trained with diverse data that may be biased and, consequently, may generate biased responses.

Back to the task on focus, in this article, we leverage summarization – a process of reducing a long text into a shorter version that preserves important information for understanding the original text^{5,11}. We aim to evaluate the LLM's ability to summarize WhatsApp (full) dialogues between patients and a healthcare team. Ideally, such summaries should capture the main characteristics of the dialogues, be concise, non-redundant, and contain truthful information.

Specifically, we propose evaluating open-source and publicly available LLMs in the task of summarizing patient messages, aiding healthcare professionals with patient-oriented, concise, quicker-to-read information to assist in responding. Moreover, such a summarization should enable healthcare teams to scale personalized care management by tailoring approaches to diverse patient needs, serving as a foundational step toward developing a digital primary care AI agent. To evaluate the feasibility of such an approach, we utilize data from real environments, including real (anonymized) interactions between patients and healthcare teams, and conduct an evaluation of these summaries with the support of volunteers from a real healthcare team.

A few studies have been conducted on summarizing clinical data. Particularly, in⁴, the authors reviewed publications on data summarization in the clinical context. The summarized data include mainly patient's electronic health records. The authors also discuss the importance of such summaries for helping tasks related to intensive care units, surgery, diagnostics, hospital care, chronic disease monitoring, oncology, drug prescription, psychotherapy, etc. Our work differs from⁴ in two main aspects: (1) data source – we explore data coming from an instant messaging system, particularly input messages from dialogues between patients and the healthcare team; such messages are usually noisy as we shall demonstrate in our analyses, potentially impacting summarization quality; and (2) application domain – our main purpose is to assist a healthcare team in responding to new patient messages, ensuring personalized, humanistic, and contextually-appropriate responses.

In sum, the main and novel contribution of this work is the evaluation of the ability of two large language models (LLMs), LLaMA 3 and Qwen 2, to summarize real-world WhatsApp dialogues between patients and healthcare teams based on real-world, noisy, and informal non-English (Portuguese) data. The study highlights the potential of LLMs to improve digital healthcare communication, particularly in resource-constrained settings like Brazil, where access to primary care is limited.

Methods

Dataset and preprocessing

We conducted this research at the Database Laboratory of Universidade Federal de Minas Gerais. The research project, including data and participation of volunteers in qualitative evaluations, was reviewed and approved by the Ethics Committee of the Federal University of Minas Gerais under registration CAAE 80632524.4.0000.5149, in accordance with the Brazilian Resolution CNS 446/12.

This study gathered WhatsApp messages from (patient, healthcare team) dialogues from October 27, 2021, until January 8, 2024, from Ana Health's cloud-based contact center. This dataset contains 207,040 messages written in Portuguese, and is available upon request from the authors. We confirm that all experiments were

conducted in accordance with the relevant guidelines and regulations, and approved by the Ethics Committee of the Federal University of Minas Gerais. We also inform that consent for the use of the dataset was obtained from the legal guardian.

Ana Health is a company that provides a digital primary care service to look after people's physical and mental health. Through a subscription plan, members (patients) gain unlimited access to Ana Health's multidisciplinary team, including general practitioners, psychologists, nurses, and gerontologists, who proactively guide them through a structured healthcare journey, complemented by 24/7 support via text messaging and phone or video calls. The company aims to scale primary care through a WhatsApp-first experience driven by a personalized care management system powered by artificial intelligence. Ana Health provided us with such messages in an anonymized way, ensuring that all identifiable information, such as names, emails, phone numbers, and URLs, was removed and replaced with a unique code to safeguard data privacy.

Regarding preprocessing, we removed special characters (tab, new lines, and duplicated white spaces) and filtered out template messages – standardized Ana Health messages to its patients – resulting in 202,326 messages and 1,863 dialogues.

Analyzing message quality

Before summarizing the dialogues, we analyze the message's textual quality to understand data limits and the potential impact of data quality on the final results. Low-quality texts can lead LLMs to generate incoherent or inaccurate summaries or summaries that do not contain the text's main information. In addition, spelling or grammatical errors may be interpreted by LLMs as unknown or inappropriate words or concepts. We based our analysis on the methodology proposed by Hasan et al.¹². We analyzed three textual quality dimensions: (1) size, (2) readability, and (3) correctness.

Size

Total number of characters, words and sentences. According to Hasan et al.¹², messages of higher quality are usually neither too short nor too long. We also define long messages as those exceeding the average length by at least ten words, and short messages as those containing five words fewer than the average message length or less, similarly to¹².

Readability

We used the Flesch-Kincaid grade level¹³, which generates an overall grade evaluating the complexity of a text, according to Eq. 1.

$$0.39 \times \frac{\text{total_words}}{\text{total_sentences}} + 11.8 \times \frac{\text{total_syllables}}{\text{total_words}} - 15.59 \quad (1)$$

As results, we have a score according to American education levels. The Flesch-Kincaid grade level considers sentences with many words or words containing many syllables more difficult to read. The Flesch-Kincaid Grade Level is a well-established readability metric that estimates the minimum education level required to comprehend a given text. In our analysis, we did not use the Flesch-Kincaid score to assess whether the texts were well-written per se. Rather, our objective was to perform a comparative analysis of the readability levels of texts authored by patients versus those written by the healthcare team. Although the Flesch-Kincaid grade level was originally developed to indicate the grade level required to understand a document written in English, there exist studies^{14,15} that adapt it to evaluate Portuguese texts. Those works show a high correlation between the values obtained by the original and adapted formulas. Thus, we may use the original Flesch-Kincaid grade level to compare documents written in the same language, as we will do in this work.

Correctness

We calculate the proportion of words in the messages that are present in a predefined Portuguese dictionary. We use the Portuguese dictionary from the package *br.ispell* (<https://github.com/fititnt/br.ispell-dicionario-portugues-s-brasileiro>).

Summarizing the dialogues

As mentioned, our main goal in this work is to evaluate the LLMs capability in summarizing (patient, healthcare team) dialogues. Ideally, such summaries should capture the main dialogue aspects and include no wrongful information. For the sake of reproducibility and transparency, we focus our study on publicly available open-source LLMs that can be run locally, ensuring data privacy, as we do not send any private patient data to remote closed LLMs such as GPT. In our experiments, we evaluated Qwen 2 (Qwen2-7B-Instruct) and LLaMA 3 (Meta-Llama-3-8B-Instruct) as LLMs to summarize dialogues. We selected LLaMA 3 and Qwen 2 based on several key considerations:

1. Open-Source Availability and Data Privacy: Both models are open-source, allowing for local deployment without reliance on external cloud infrastructure. This aspect is crucial for our application, as it enables the system to operate without transmitting sensitive user data over the internet, thereby mitigating potential risks related to data privacy and security, especially relevant when considering future production-level adoption by the partner company.
2. Empirical Performance: LLaMA 3 and Qwen 2 have demonstrated state-of-the-art performance across a range of natural language processing tasks, including question answering, classification, code generation, text rewriting, and reasoning. Public benchmarks and evaluations (e.g., Meta AI blog on LLaMA 3 and Hug-

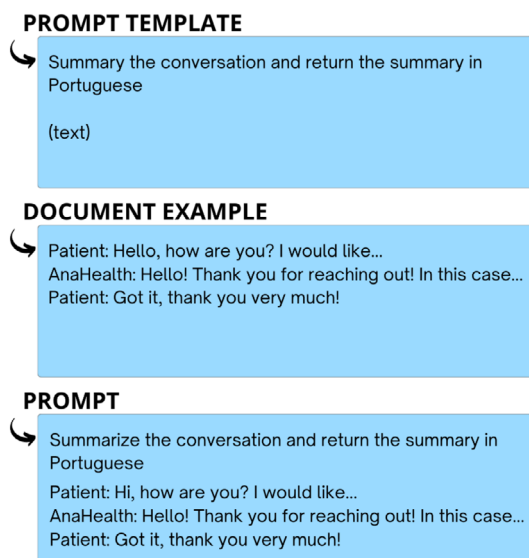


Fig. 1. Example of prompt.

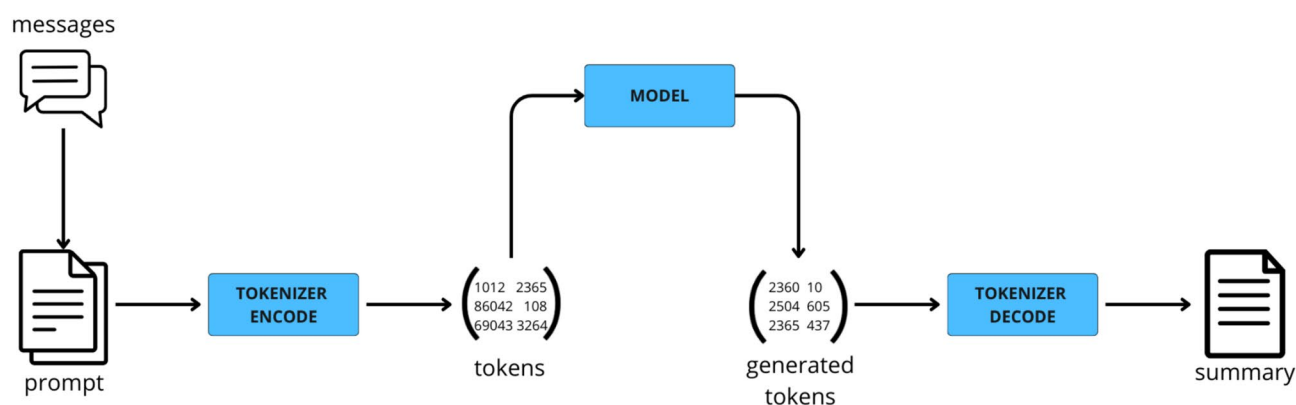


Fig. 2. Summarization process using LLM.

ging Face model card for Qwen2-7B) support the superiority of these models compared to other open-source alternatives.

3. **Model Size and Computational Efficiency:** We employed the 8-billion-parameter version of LLaMA 3 and the 7-billion-parameter version of Qwen 2. These models strike a balance between performance and computational cost, allowing us to run them on standard local hardware without requiring specialized or high-cost infrastructure.

For the summarization process, all messages from a (patient and healthcare team) dialogue are grouped together to generate a single document (possibly a long one). More formally, for each patient p_i belonging to a set of patients $\{p_1, p_2, \dots, p_n\}$, we produce a document d_i with all messages related to p_i . A summarization method receives a document d_i for p_i as input and must produce a summary s_i capturing the main aspects and topics discussed in d_i .

For using LLMs as a summarization method, we developed a prompt, illustrated in Fig. 1. This prompt contains instructions for performing the summarization task. The text to be processed by a LLM must be tokenized, i.e., it is broken down into smaller subword units known as tokens, resulting in a sequence of tokens. Each subword is identified numerically. The LLM processes the sequence of tokens and generates a different sequence to create the respective summaries. Figure 2 depicts the full summarization process.

LLMs can only receive a limited number of tokens as input. In our work, as we deal with messages in chronological order, we use the last 5,000 tokens from a dialogue as input to the LLMs, as the last messages of the dialogue may be more important to respond to a current patient message.

Evaluating the summaries

To evaluate the LLM-generated summaries, we performed an A/B test. We conducted the evaluation of the generated summaries using volunteers from Ana Health's healthcare team. It is important to stress that the participants who evaluated the summaries were not involved in the dialogues with the patients to avoid any bias that may come from previous knowledge of the patient.

For evaluation, we did not exploit traditional summarization metrics, such as ROUGE¹⁶, BLEU¹⁷, NIST¹⁸, and METEOR¹⁹, which are based on text overlap, or BERTScore²⁰ and BARTScore²¹, which are similarity-based metrics, as these metrics require ground-truth information, which is not available in our dataset. In other words, we do not know what would be the “perfect summary”. Some recent works also analyze the use of LLMs as summary evaluators^{22–24}. However, our goal is not to analyze the ability of LLMs to evaluate the quality of summaries. Instead, we focus on evaluating their performance in generating useful summaries about patients, when facing noisy and informal data, with the goal of helping a healthcare professional to appropriately respond to the patient's messages. Thus, we conducted an evaluation by humans, in particular potential real users of the generated summaries. This constitutes one main contribution of our work.

Evaluation criteria for the summaries

We elaborated a survey in which each participant had to read messages from the original entire dialogue (conversation) and, next, rate each summary using a 5-point Likert scale (1 - strongly disagree, 5 - strongly agree)²² considering four perspectives:

- Coverage: The summary covers all important and relevant aspects of the full conversation;
- Relevance: All aspects of the summary are relevant to the context of the messages, i.e., the summary does not contain irrelevant portions;
- Redundancy: The summary is non-redundant, i.e., it does not contain repetitive information;
- Veracity: The summary is truthful (does not contain wrongful information).

After reviewing the literature^{3,25} and discussing the metrics with the Ana Health healthcare team, we focus our assessment on these perspectives. A supplementary file contains an example of our survey.

Participants

As volunteers/participants to evaluate the summaries, we had five physicians and 19 psychologists. We selected 24 healthcare professionals who are responsible for welcoming patients, conducting triage based on patient-reported information, and maintaining an accurate understanding of each patient's history and most recent interactions with the company. It is also important to highlight that the literature indicates that usability testing with more than 15 users is typically sufficient to identify the vast majority – if not all – of the major issues²⁶. We distributed the survey among participants to ensure a balanced representation of professional types. As additional information, the participants are predominantly women, with approximately 22% being men. Each summary was evaluated by two women and one man, or by three women.

Evaluation protocol

In the survey, we applied the same evaluation protocol to all participants (both external and internal) to minimize bias in our results. We estimated that, to maintain an evaluation time of around 15 minutes, each participant should evaluate two summaries to avoid fatigue and ensure effective participation throughout the study, including the external psychologist. Several studies^{27,28} in the area of research methodologies indicate that surveys lasting less than 15 minutes tend to have a higher response rate, fewer dropouts midway through the questionnaire and better quality responses. For each summary, participants first read the full dialogue, then read the summary, and finally rate the four perspectives based on the Likert scale.

Before randomly selecting the dialogues to be summarized in our experiment, we first filtered dialogues with a token count between 1,000 and 5,000. This step ensures that the dialogues will be neither truncated by the LLM nor too brief to contain insufficient information for generating a summary. Next, we randomly selected eight full dialogues and provided them to both LLMs — LLaMA 3 and Qwen 2. Consequently, we obtained sixteen summaries for evaluation, with each summary assessed by three different participants.

Analyzing survey data

To analyze the survey data, we first analyze the distributions of responses in each perspective, aiming to obtain an initial understanding of the behavior of the participants' evaluations for each LLM. Then we use the numerical scale (1 - strongly disagree – 5 - strongly agree) to analyze the mean values and their standard deviations to obtain a quantitative interpretation of the LLMs effectiveness.

We also calculated the Kappa-Cohen coefficient (k) to measure the agreement between two participants. It is useful for assessing the consistency of the obtained responses. We applied this metric exclusively to evaluating the same perspective on the same summary by two participants. After calculating the agreement for all possible pairs of responses, we computed the average to determine an overall agreement value.

$$k = \frac{p_o - p_e}{1 - p_e} \quad (2)$$

where p_o represents the observed agreement ratio, reflecting the empirical probability of agreement on the answers provided for any given statement and p_e represents the expected agreement, assuming that the evaluators respond randomly. By incorporating p_e , we account for the possibility of agreement occurring by chance.

We conducted a final qualitative analysis of the generated summaries in our case study using the perspective of a professional focused on the humanized reception of patients in the context of the psychotherapeutic process.

Data privacy

This study was conducted in accordance with the highest ethical standards and received prior approval from an independent institutional ethics committee. Specific attention was given to the issues of data anonymization, privacy preservation, and the ethical implications of processing patient-related communication.

The dataset used in our study was anonymized prior to our access, with all identifiable information removed by the partnering company in compliance with their internal data governance policies. We did not perform any data collection ourselves. Instead, we received the pre-processed data in anonymized form, with no access to any personal identifiers such as names, contact details, or healthcare identification numbers.

Furthermore, all data processing, including the summarization task, was conducted locally using on-premise infrastructure. At no stage were the data or derived outputs transmitted to cloud-based services or external servers. This approach ensures that the risk of data exposure through insecure communication channels is eliminated.

To strengthen our commitment to confidentiality, all members of the research team signed a legally binding Non-Disclosure Agreement (NDA) with the data-providing entity, ensuring that no part of the data or results could be used or disclosed outside the scope of the approved study.

Regarding the operational deployment of the system, it is important to clarify that only authorized healthcare professionals who are already granted access to the original patient communications within the company's clinical workflow will be able to access the summaries generated by the system. These professionals are, by definition, bound by professional confidentiality and ethical obligations, and already possess access to more detailed and potentially identifiable information than what is present in the summaries. Therefore, the summarization component does not introduce any additional privacy risks, nor does it expand the circle of data access beyond what is already sanctioned by the healthcare provider.

Our study applied state-of-the-art anonymization techniques, removed direct identifiers, and ensured that all work was conducted under strict legal and ethical safeguards. These measures significantly mitigate the risk of deanonymization and are consistent with ethical research practices and data protection regulations such as national data protection laws applicable to our context.

Results

Evaluating the quality of the messages

In this section, we discuss the quality of the messages regarding three dimensions: size, readability, and correctness.

Size

Figure 3 shows the distribution of the size of messages in words, characters, and phrases, considering all messages exchanged by patients and the healthcare team, whereas Table 1 shows average, maximum and minimum values of characters, words, and phrases - along with their quartile boundaries - considering all messages. Up to 75% of the messages have up to 135 characters, 25 words, and 4 phrases, respectively.

We found 47,443 (23.45%) long messages and 112,971 (55.84%) short messages. That is, most of the messages are short, while almost a quarter are long.

Readability

We utilized the Flesch-Kincaid grade level for the readability analysis. Before applying the Flesch-Kincaid grade level, we grouped all messages to and from the same patient, resulting in three types of "documents": (i) all messages sent by the healthcare team to a patient p_i and all the messages sent by p_i to the healthcare team, and (iii) the union of both, i.e., all messages from or to each patient (all messages per patient). As discussed above, we separated the analysis according to the direction of the message flow. Figure 4 and Table 2 show the results.

Comparing the grade levels of the groups of messages sent by the healthcare team with those of the groups of messages sent by patients, we observed higher average values and quartile limits for those sent by the healthcare team. This indicates that, on average, texts produced by the healthcare team are more complex and less simplistic than those produced by patients (they require a higher educational level to be read). The difference is noteworthy and should be considered in the development of future methodologies that rely on this type of data. The distribution of grade levels for the messages sent by patients has a wider range of values, meaning that patient messages have a higher variability in terms of quality.

Correctness

For this analysis, we used a Portuguese dictionary from *br.ispell* to compare the proportion of correctly spelled words in messages sent by patients with those sent by the healthcare team. The results are shown in Fig. 5 and Table 3.

Notice that healthcare team messages exhibit a higher level of correctness compared with those sent by patients. Patients' messages have very low correctness, indicating poor writing quality, containing a lot of typos, misspellings, abbreviations, unknown words, etc.

Evaluating the summarization task

In this section, we discuss the assessment of the summarization task. Before describing the results, we present some statistics about the lengths (word counts) of our sample of dialogues and their corresponding summaries generated by LLaMA 3 and Qwen 2. Table 4 presents the word counts of the original dialogues and their

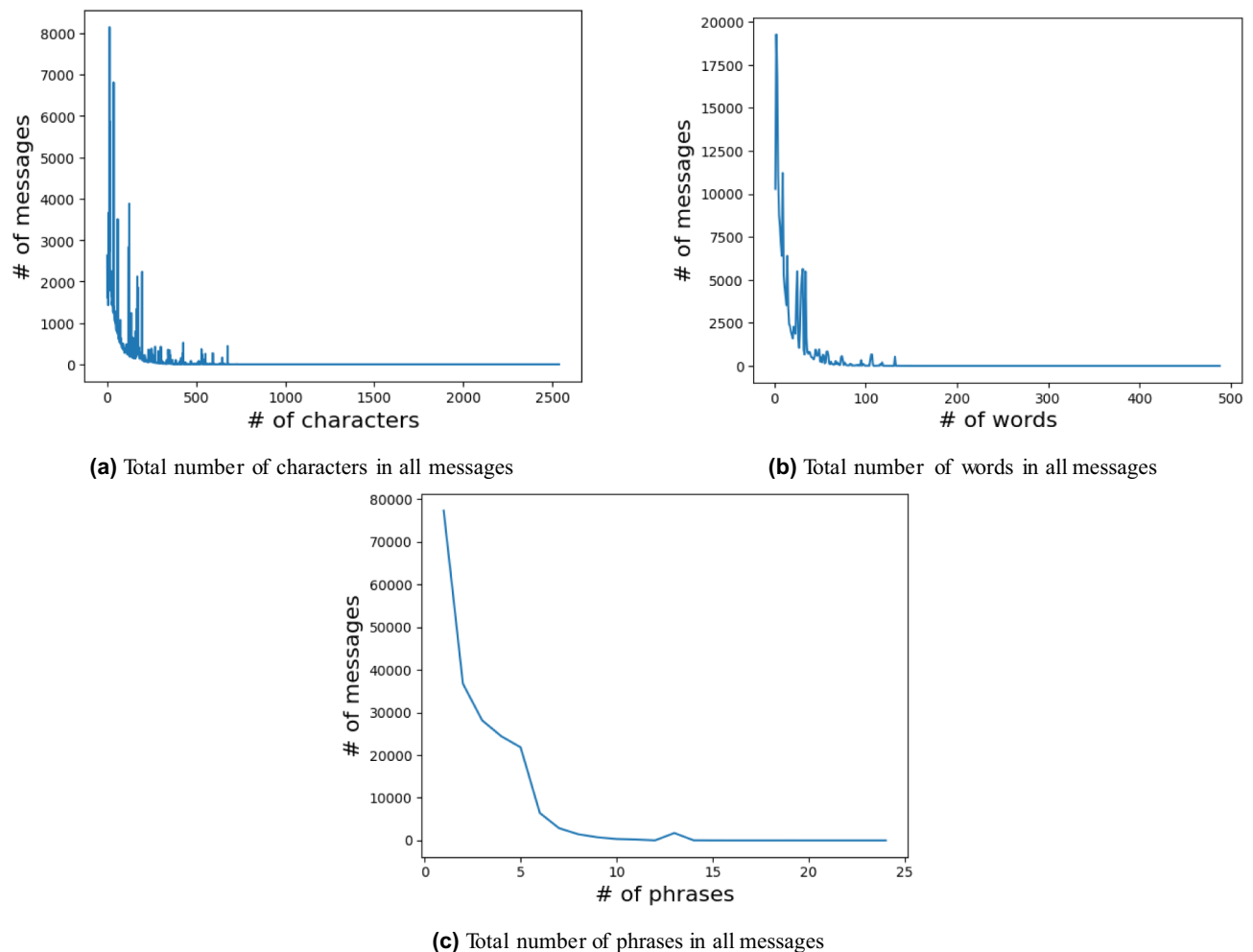


Fig. 3. Distribution of the size of **a** Words, **b** Characters and **c** Phrases in all messages (202,326 messages) exchanged between patients and the healthcare team. The majority of the messages has few characters, words and phrases.

	Average (SD)	Minimum	Maximum	Q1	Q2	Q3
Characters	91.45 (109.49)	1	2,538	17	48	135
Words	17.47 (19.95)	1	488	4	10	25
Phrases	2.71 (2.01)	1	24	1	2	4

Table 1. Size - average (standard deviation), minimum and maximum values, and quartile boundaries (Q1, Q2 and Q3) from 202,326 messages. Up to 75% of the messages have 135 characters, 25 words, and 4 phrases.

corresponding summaries generated by LLaMA 3 and Qwen 2. It is worth noting that there is no clear correlation between the length of the input dialogues and the length of the summaries generated by the language models. On average, the summaries generated by LLaMA 3 and Qwen 2 contain approximately 137 and 147 words, respectively, with standard deviations of 59 and 41 words. This indicates that the models produce summaries with lengths around 150 words, regardless of the length of the original dialogue.

We conducted our assessment of the summarization task through a survey with 24 participants, including five physicians and 18 psychologists from Ana Health's healthcare team, and one external psychologist. To analyze the survey data, we use two approaches. First, we analyze the distributions of the scores the participants gave for the evaluated summaries. Second, we employ statistical metrics to obtain a quantitative interpretation of the LLMs' effectiveness.

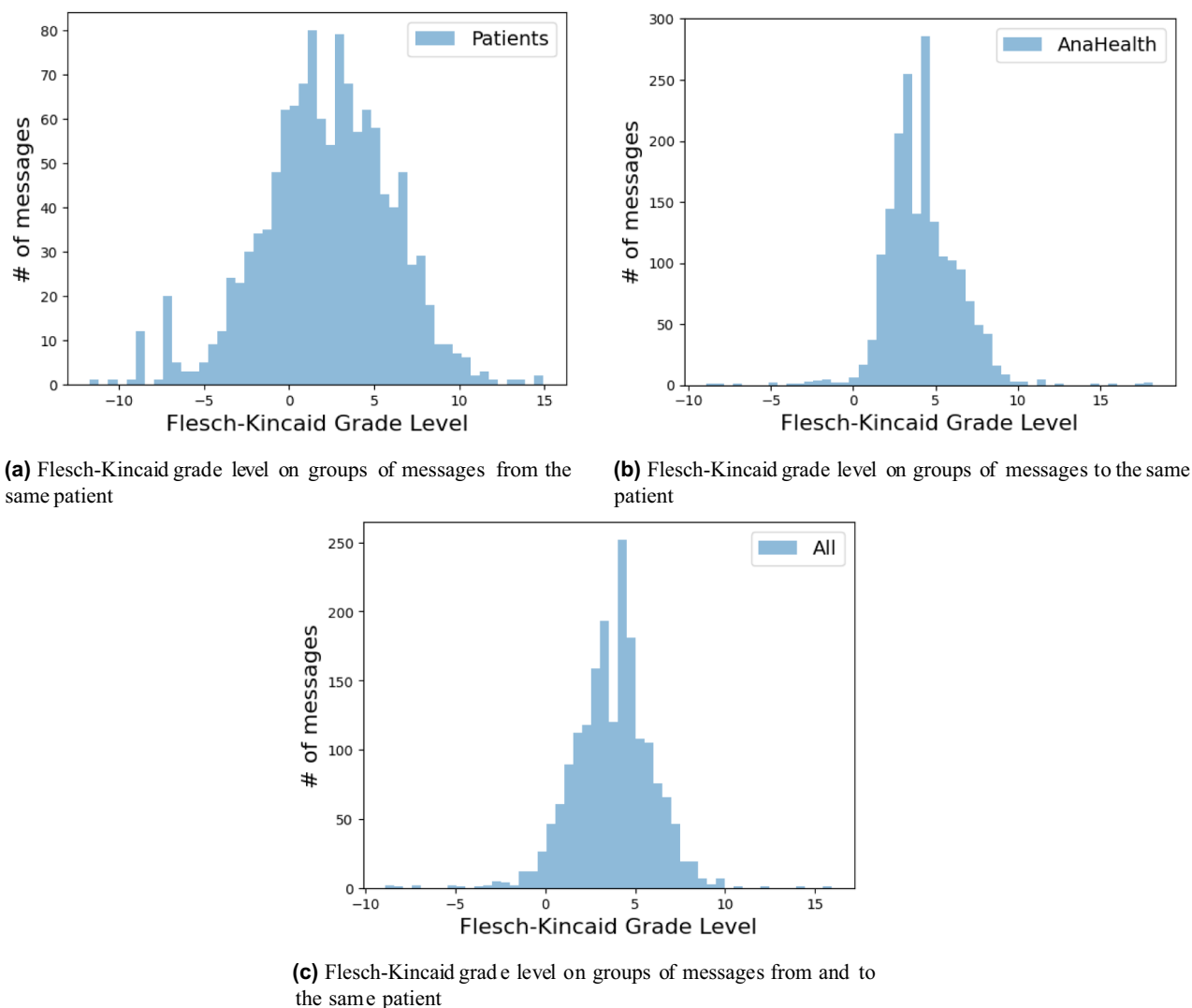


Fig. 4. Distribution of the Flesch-Kincaid grade level for **a** Groups of messages sent by each patient to the healthcare team, and **b** Groups of messages sent by the healthcare team to each patient and **c** Groups of messages from or to the same patient. The messages sent by patients have a wider range of grade levels, meaning that patient messages are spread across a broader grade level interval compared to the healthcare team.

Grouping criterion	# of groups	Flesch-Kincaid grade level					
		Average (SD)	Minimum	Maximum	Q1	Q2	Q3
From or to the same patient	1,863	3.72 (2.20)	-8.91	15.95	2.36	3.88	4.99
To the same patient	1,859	4.18 (2.16)	-8.91	18.20	2.82	4.08	5.35
From the same patient	1,224	2.18 (3.85)	-11.69	14.95	-0.18	2.32	4.82

Table 2. Readability - total number of groups of messages, average (standard deviation), minimum and maximum Flesch-Kincaid grade levels, and quartile boundaries (Q1, Q2, and Q3). High average values and quartile limits for the groups of messages sent by the healthcare team indicate that the messages sent by the healthcare team require a higher grade level.

Distribution of responses for each perspective

We conducted this analysis individually for each evaluated perspective, enabling us to identify response patterns and directly compare the LLMs. Figure 6 illustrates the distribution of responses for each perspective when using LLaMA 3 in the summarization task.

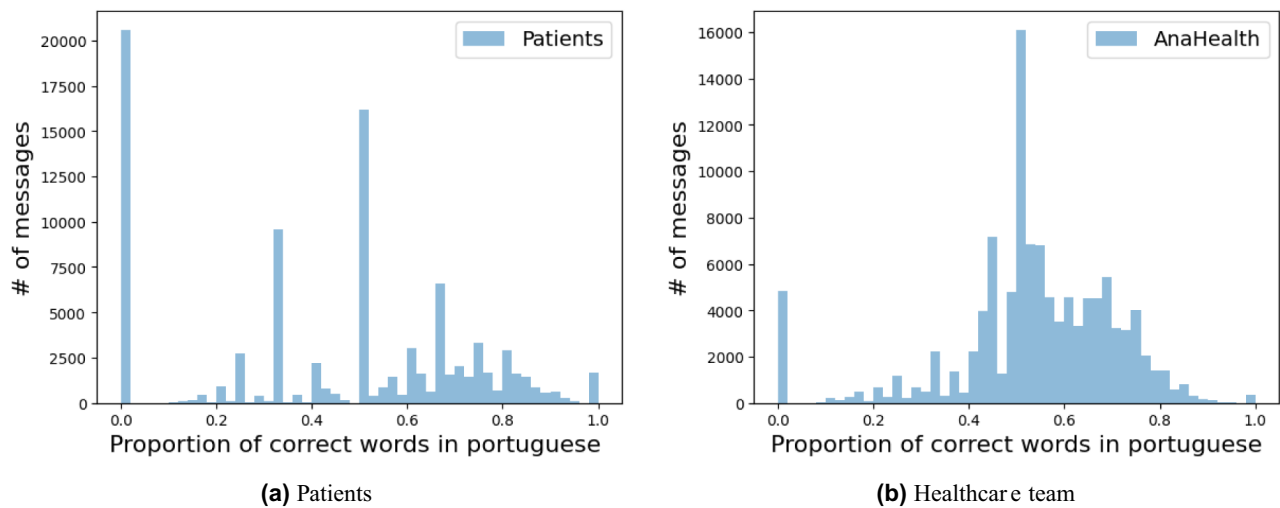


Fig. 5. Distribution of correctly spelled words in messages sent by **a** Patients and **b** Healthcare team. The messages sent by the healthcare team have a higher level of correctness compared with those sent by patients. 202,326 total messages, 111,011 healthcare team’s messages and 91,315 patients’ messages..

Source	Average (SD)	Minimum	Maximum	Q1	Q2	Q3
All messages	0.49 (0.24)	0.00	1.00	0.40	0.52	0.67
Healthcare team’s messages	0.54 (0.18)	0.00	1.00	0.47	0.54	0.66
Patients’ messages	0.44 (0.29)	0.00	1.00	0.25	0.50	0.67

Table 3. Correctness - average (standard deviation), minimum and maximum values, and quartile boundaries (Q1, Q2, and Q3). 202,326 total messages, 111,011 healthcare team’s messages and 91,315 patients’ messages..

Dialogue	# of words		
	Dialogue	LLaMA Summary	Qwen summary
1	967	102	195
2	2069	208	202
3	1254	85	164
4	1458	220	167
5	1422	106	138
6	2300	67	124
7	2767	122	107
8	2659	189	85

Table 4. Word counts of dialogues and corresponding summaries generated by LLaMA 3 and Qwen 2.

Notice that, across all perspectives, positive responses — “strongly agree” and “agree” combined — account for more than 50% of the total responses. More specifically, the percentages are 62.5%, 58.3%, 54.2% and 70.9%, respectively, for coverage, relevance, redundancy, and veracity regarding LLaMA 3 summaries. Furthermore, “strongly disagree” responses are a minority across all perspectives and are entirely absent for model’s coverage. In sum, volunteers considered LLaMA 3 summaries predominantly satisfactory in basically all perspectives.

For Qwen 2, Fig. 7 also shows a predominance of positive responses, which, as for LLaMA 3, account for more than 50% of the total. That is, the percentages of positive responses are 54.2%, 58.3%, 70.8% and 54.1%, for coverage, relevance, redundancy, and veracity, respectively, also suggesting a good summarization capability for this LLM. However, “strongly disagree” responses for Qwen 2 are significantly more frequent than those for LLaMA 3, exceeding 10% across all perspectives.

Analyzing statistical metrics

Table 5 shows the average rates for each perspective organized by LLM. All values are above 3, indicating a tendency for positive results and suggesting a good model (average) effectiveness in both cases. For coverage and relevance, LLaMA 3 obtains average values slightly higher than those obtained by Qwen 2. Conversely, Qwen

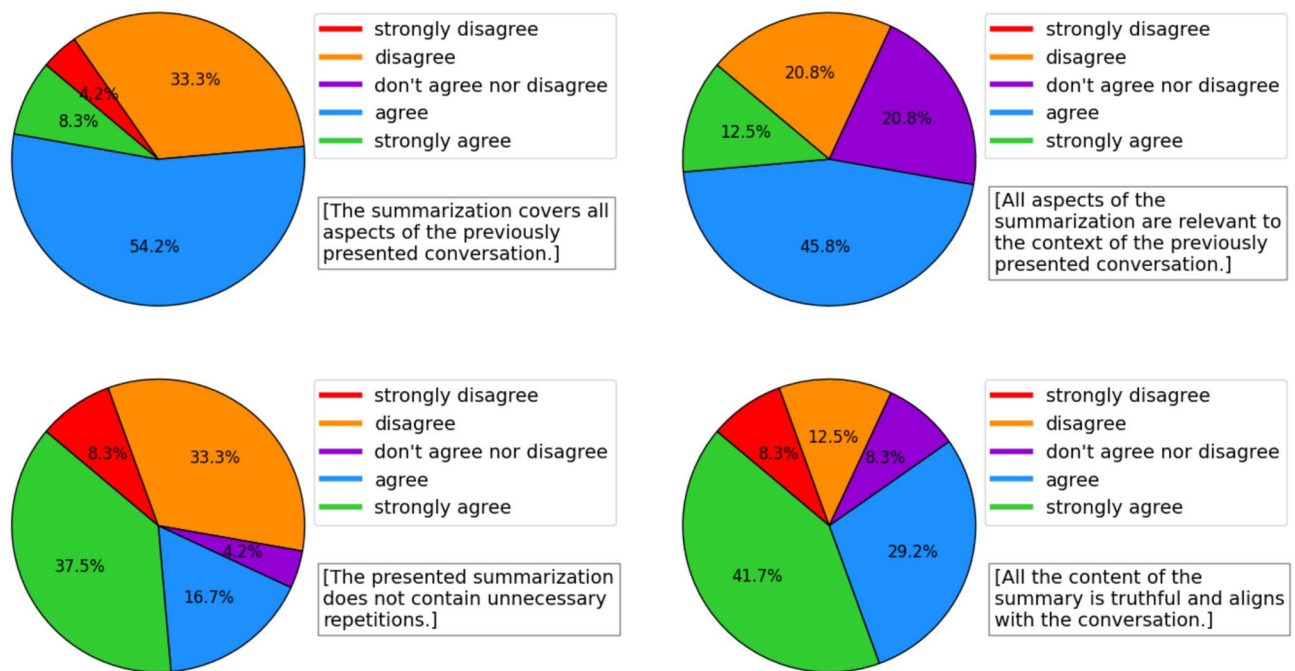


Fig. 6. Distribution of scores for each perspective (24 assessments per perspective) evaluating summaries generated by LLaMA 3. Positive responses — “agree” and “strongly agree” — are more than 50% of the total responses.

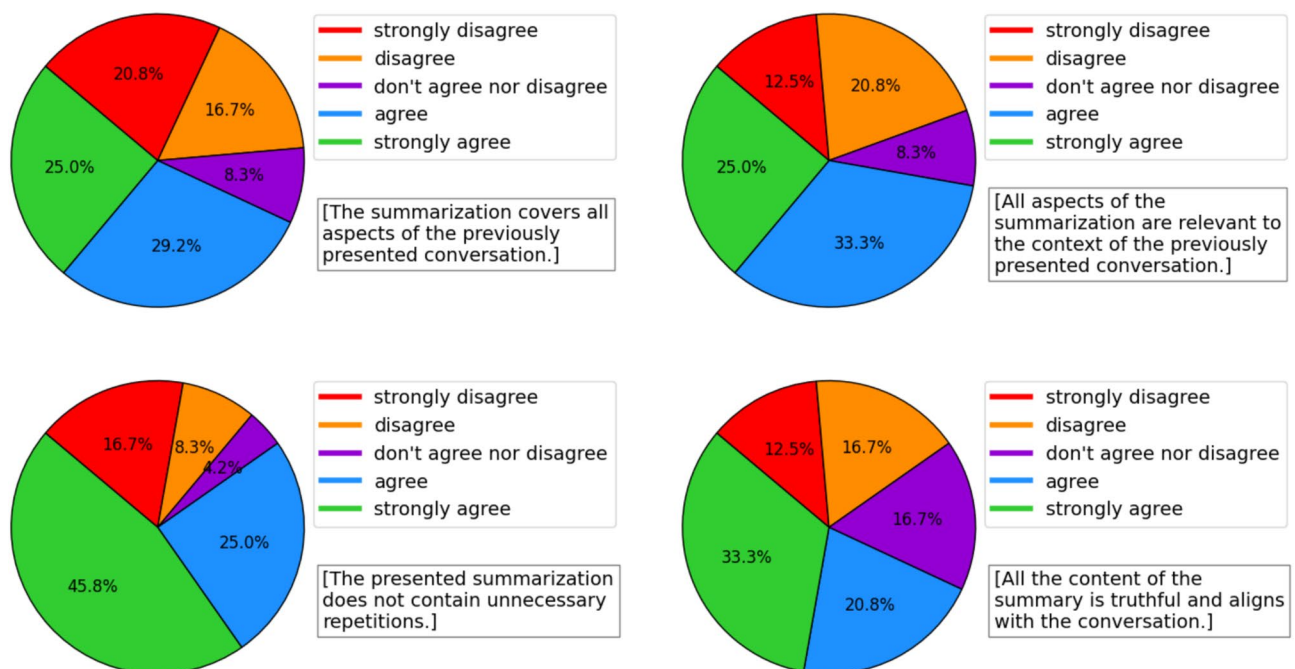


Fig. 7. Distribution of scores for each perspective (24 assessments per perspective) evaluating summaries generated by Qwen 2. Positive responses – “agree” and “strongly agree” – are more than 50% of the total responses.

2 stands out in (lack of) redundancy concerning LLaMA 3. The opposite is true for veracity, indicating that, on average, LLaMA 3 is more truthful than Qwen 2.

We also note that, for all perspectives, the standard deviations obtained using LLaMA 3 are smaller, indicating less data dispersion. This suggests less variation in the participants’ responses, pointing to a higher level of consensus, which is crucial for selecting a model. We elaborate further on that next.

Perspective	LLaMA 3	Qwen 2
Coverage	3.29 (1.16)	3.21 (1.53)
Relevance	3.50 (0.98)	3.38 (1.41)
Redundancy	3.42 (1.50)	3.75 (1.54)
Veracity	3.83 (1.34)	3.46 (1.44)

Table 5. Average values for each perspective per LLM. Standard deviation in parentheses.

Summary	LLaMA 3	Summary	Qwen 2
s_1	0.116	s_8	0.313
s_2	0.036	s_7	-0.022
s_3	0.283	s_6	0.056
s_4	0.087	s_5	-0.132

Table 6. The Kappa-Cohen average values for each LLM per summary.

Finally, we did not observe any correlation between the lengths of dialogues and summaries and the qualitative assessments conducted in our study. These findings suggest that summary length, within the observed range, does not significantly influence the evaluated perspectives.

Kappa-Cohen (agreement between participants)

Since the Kappa-Cohen coefficient is only meaningful when calculated in evaluations carried out on the same object (summary), we selected four summaries generated by LLaMA 3, represented in Table 6 as s_1 , s_2 , s_3 and s_4 , and four summaries generated by Qwen 2, represented as s_5 , s_6 , s_7 and s_8 , totaling eight summaries. Since three participants evaluated each summary, we have three Kappa-Cohen values for each summary and show the average value of these three values in Table 6.

Kappa-Cohen values range from -1 to 1, with values lower than 0 indicating that the agreement is worse than what would be expected by chance. We observed such a situation in the evaluations of summaries s_5 and s_7 generated by Qwen 2. Values between 0.0 and 0.2 indicate weak agreement, which we observe in various results. Values between 0.2 and 0.4 indicate reasonable agreement, as occurs in s_3 and s_8 generated by LLaMA 3 and Qwen 2, respectively.

Overall, when analyzing the values obtained by applying the Kappa-Cohen coefficient, we note that the agreement values were generally low, with LLaMA 3 results being more consistent than Qwen 2's. However, it is important to stress that since this metric considers only total agreement (same response), it is very sensitive to the fact that there are five possible responses. In other words, there is no consideration of the *degree of gradation* of the responses, being the disagreement between "agree" and "strongly agree" considered the same as between "strongly agree" and "strongly disagree", for example.

External Qualitative Analysis

Aiming to have a view of a newly integrated professional into the healthcare team, we asked the external participant – in this case, a psychologist who was not part of Ana Health's healthcare team – to assess the usefulness of the summaries in helping recently enrolled collaborators understand the context when patients send new messages. In summary, taking two dialogues as input, we used Qwen 2 and LLaMA 3 to generate two summaries for each dialogue. The participant evaluated the summaries regarding clarity, cohesion, and usefulness of the summaries without knowing which LLMs were used.

When evaluating the summaries generated by the first LLM (Qwen 2), she found the first summary satisfactory, as it accurately reflected the data discussed in the conversation between the patient and the healthcare team. However, the participant considered the second summary, also produced by Qwen 2, unsatisfactory since it lacked important data necessary for a complete understanding of the case, making it impossible to provide an attentive and efficient service.

Furthermore, according to the participant, the second summary lacked the human aspect and failed to meet the requirements for understanding the case, given that the model failed to provide crucial data, such as the fact that the patient was recovering from COVID, for instance. Furthermore, the summary contained details about how the patient had expressed gratitude for the care, which was considered unimportant to understanding the patient's condition.

Regarding the summaries generated using the second LLM (LLaMA 3), the participant considered that both summaries maintained a certain standard of quality as well as efficiently addressed the content of the conversation. In this sense, the four previously requested observation points were met, as all aspects of the conversation were well summarized, had the necessary relevance for understanding the case, were objective and without irrelevant repetitions, and did not present false or dubious content.

According to the participant, summaries with positive evaluations from the perspectives evaluated can help a new member understand the context of new messages and thus formulate contextualized and personalized responses to the patient with agility. She also informs that the ability to understand each individual's context, as

managing the care reception process is inherently personal and non-transferable. “Each person is unique and requires a tailored approach to her healthcare. Personalized reception, adapted to each individual’s specific needs and characteristics, is essential for creating a truly humanized and effective experience”. Indeed, time is often a critical factor in the integration process. Faster and more accurate access to information improves the efficiency of this process²⁹.

Discussion

Previous work on summarizing clinical data has not delved deeply into the quality of the input data, which, in our case, as we shall see, can be very poor and informal. This analysis itself is a contribution of this work, as it puts a high burden on the LLM-based summarizers. On top of that, we have the fact that the dialogues are in Portuguese. Indeed, while being one of the most spoken languages globally, Portuguese has historically been underrepresented in training many large language models (LLMs) compared to languages like English, Spanish, or Chinese. This can be attributed to several factors, including the availability of high-quality training data, the focus of research and development efforts, and the predominant use of English in the tech industry³⁰.

Message quality

Regarding the size dimension, we observed that most messages were considered short (56.12%) and around 23.58% were considered long. Short texts may lack information or contain incomplete information, while long texts may have redundancies, which may lead LLMs to not focus on the main parts of the text. Both issues may affect summary quality.

The average Flesch-Kincaid grade levels for the healthcare team and patients suggest that a higher grade level may be required to understand healthcare team messages compared with those from patients. Regarding patients, we observe that their messages have a wider range of values for the Flesch-Kincaid grade level than those from the healthcare team. Although this may indicate a potential diversity of patients in terms of education level, several other factors may also influence these results, such as the level of detail needed to talk about certain subjects and the use of audio to explain more complex subjects, among others.

Regarding correctness, most words in the messages sent by the healthcare team are found in a standard dictionary, while this trend is not found for patient messages. Indeed, the proportion of “correct” words in patient messages is very low, around 44% on average. However, this low correctness level may be taken in context and be more carefully analyzed as the language usage of WhatsApp users has its own distinct terms, such as “vc” (abbreviation of “você” – you in Portuguese), “tbm” (abbreviation of “também” – also in Portuguese), “pq” (abbreviation of “porque” – why or because in Portuguese), in which, from the perspective of the users are common and seen as correct. In any case, the high occurrence of incorrect or non-standard words may lead LLMs to ignore or misinterpret them in the summaries, causing omissions of important parts from the original text or generation of wrong information.

Overall, considering the dimensions evaluated of size, readability, and correctness, there are clues of low quality in patients’ messages. This (poor) data quality in our dataset may pose significant challenges for summarization tasks using LLMs.

Summarization quality

We evaluated the summaries from four perspectives, coverage, relevance, redundancy and veracity, through an online survey. Both LLMs performed well from all four perspectives. More than 50% of the responses are “strongly agree” and “agree” in all perspectives and in both LLMs. This shows the ability of the LLMs to generate satisfactory summaries despite the quality of the original text. The few “strongly disagree” responses obtained in our analysis (less than 10%) are concentrated on Qwen 2 summaries, being completely absent in LLaMA 3’s coverage.

Comparing the results of LLaMA 3 with those of Qwen 2, considering “strongly agree” and “agree” responses, we notice that, for coverage and veracity, LLaMA 3’s summaries are slightly superior to those produced by Qwen 2. Both LLMs tie under the relevance perspective and the Qwen 2 summaries are slightly superior regarding (lack) of redundancy. Overall, considering all evaluation criteria, LLaMA 3’s summarization performance in our application is slightly superior to Qwen 2’s.

In an external qualitative analysis, viewed from the perspective of a newly integrated professional into the healthcare team, the professional observed that the summaries produced by LLaMA 3 are objective, free of irrelevant repetitions and do not contain false or dubious content. On the other hand, the summaries produced by Qwen 2 do not meet the same standard. While a summary correctly reflects the content of the dialogue, the other one lacks important data included in the dialogue.

Positive evaluations from the analyzed perspectives potentially mean summaries that can effectively help team members quickly understand the context of a new message without needing to review the entire prior dialogue with the patient. This also enables team members to respond to the new message in an efficient and contextually appropriate manner. Such agility in providing clear and accurate information may help reduce the anxiety of patients awaiting a response, thus making the interaction process between patients and healthcare team members more efficient.

In short, despite the messages being written in Portuguese, some clues of limited quality of the messages and high variability in the grade level required to understand the messages, the summaries produced by both evaluated LLMs are satisfactory for the sake of the target task, with LLaMA 3 showing slight superiority.

Limitations

Our work has some limitations. First, we generate summaries using messages exclusively exchanged between patients and the Ana Health healthcare team. Our goal in this work is only to evaluate the ability of LLMs to

generate summaries. If we aim to create summaries containing complete patient information, such as their health history, preferences, and demographic details, additional data sources will be required. Second, our evaluation of the summaries was carried out with members of the Ana Health healthcare team. Perhaps people from other areas or even patients may have different opinions. Third, we evaluated the summaries using only the Likert scale. If we wanted to understand in depth the reason for each value recorded by the participant, we would need other types of responses, such as comments on the evaluated summaries. Fourth, our evaluation used messages from the dialogues of eight patients as a source. A more comprehensive analysis of LLMs summarization capability may require a larger number of dialogues. Fifth, this work did not evaluate the use of summaries in other tasks, such as their impact on responding to patient messages.

Future directions

We are currently exploring several avenues to enhance the performance and applicability of the summarization system in future work. These efforts include incorporating electronic medical records to enrich the input data, integrating audio processing to capture and utilize verbal communication, and adopting strategies that enable the separate analysis and summarization of distinct segments of the data prior to generating a comprehensive final summary.

Furthermore, in our qualitative analysis of LLMs' ability to summarize dialogues, we deliberately limited the amount of input data to ensure it remained within the context window of the models. At this stage, we consider this a reasonable constraint, as our primary goal was to assess how effectively the models could capture the essence of the most recent interactions between patients and the healthcare team – under the assumption that recent messages are more likely to reflect the patient's current health status. As part of our future work, we intend to investigate methods to address the limitations imposed by the context window size of large language models, which remains a critical factor influencing summarization quality^{5,31}.

In this work, an external psychologist evaluated two summaries according to the same protocol applied to all participants. In the future, we also intend to include more external evaluators for improving our evaluation.

Conclusions

In this article, we evaluated the ability of two large language models, LLaMA 3 and Qwen 2, to summarize dialogues between patients and healthcare teams. These dialogues, written in Portuguese, were conducted through WhatsApp. Ideally, the generated summaries should highlight important information about each patient, enabling the healthcare team to respond to new patient messages quickly and in a personalized manner.

Differently from previous work, which applied summarization in high-quality clinical data (e.g., electronic health records), we start our contributions by first assessing the quality of the dialogues to be summarized and found evidence of low data quality, especially due to informality. The primary objective of our study is to rigorously evaluate whether state-of-the-art LLMs – specifically open-source models – are capable of producing clinically useful summaries from real-world, informal, and noisy patient-provider dialogues in Portuguese, a language underrepresented in current medical NLP research.

Our contribution lies in the application and systematic evaluation of these models in a realistic healthcare communication scenario, which presents several technical and practical challenges: the dialogues are unstructured, colloquial, and often fragmented, reflecting actual communication patterns in digital health platforms; the setting involves asynchronous, chronologically evolving dialogues, requiring the models to maintain contextual coherence over multiple messages; the domain involves sensitive health-related content, where summarization errors can have non-trivial implications.

To address concerns regarding the robustness and safety of the generated summaries, we structured our evaluation around four key criteria: coverage, relevance, redundancy, and veracity. These dimensions were chosen precisely to assess potential summarization errors that could impact clinical interpretation. All evaluations were conducted manually by human annotators familiar with the healthcare domain and native in Portuguese. Our results suggest that, while both models perform reasonably well, there are notable differences in their ability to generate true and relevant summaries – issues that are critical in any health-related application.

To summarize, the novelty of our work lies in: (1) the application and comparative evaluation of LLMs in a low-resource, high-stakes domain using real-world Portuguese clinical dialogues; (2) the design of a multi-faceted evaluation framework specifically tailored to identify errors that could compromise clinical decision-making; and (3) the establishment of a foundation for future clinical validation, which will directly address utility and patient safety concerns raised by the reviewer.

In conclusion, even when processing low quality Portuguese texts, the evaluated LLMs were able to produce summaries that could significantly improve patient care. These summaries support healthcare team members in understanding the context of new messages from patients, enabling them to respond quickly and in a personalized manner, thereby enhancing patient outcomes.

From a broader perspective, our study underscores the crucial role of LLMs in addressing healthcare disparities. In countries like Brazil, where a significant portion of the population lacks access to basic healthcare and there is a shortage of health professionals, digital healthcare services are not just essential, but they represent a critical solution. Strategies, such as those discussed in this article, that can effectively process large volumes of data in underrepresented languages have the potential to make a significant social impact by helping to alleviate pressing issues.

Data Availability

The data generated or analyzed during this study are included in this article and its supplementary information files. The corresponding author is available to provide additional data regarding this manuscript upon request.

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Author contributions

Conception or design of the work: AAF, MAG, LR, and WC. Data collection: ACM, APSJ, GJ, GPS, IA, JMC, WAS, and WC. Data analysis and interpretation: AAF, ACFV, ACM, ET, GJ, IA, JMC, LR, MAG, and WC. Critical revision of the article: AAF, ACFV, ACM, APSJ, ET, GJ, GPS, IA, JMC, JMP, LR, MADs, MAG, OSN, VM, WAS, and WC. Final approval of the version to be published: AAF, ACFV, ACM, APSJ, ET, GJ, GPS, IA, JMC, JMP, LR, MADs, MAG, OSN, VM, WAS, and WC.

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Declarations

Competing interests

The authors declare no competing interests.

Human Ethics and consent to participate

This work, including data and participation of volunteers in qualitative evaluations, was reviewed and approved by the Ethics Committee of the Federal University of Minas Gerais, in accordance with the Brazilian Resolution CNS 446/12, under registration CAAE 80632524.4.0000.5149. Informed consent was obtained from all the participants.

Additional information

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